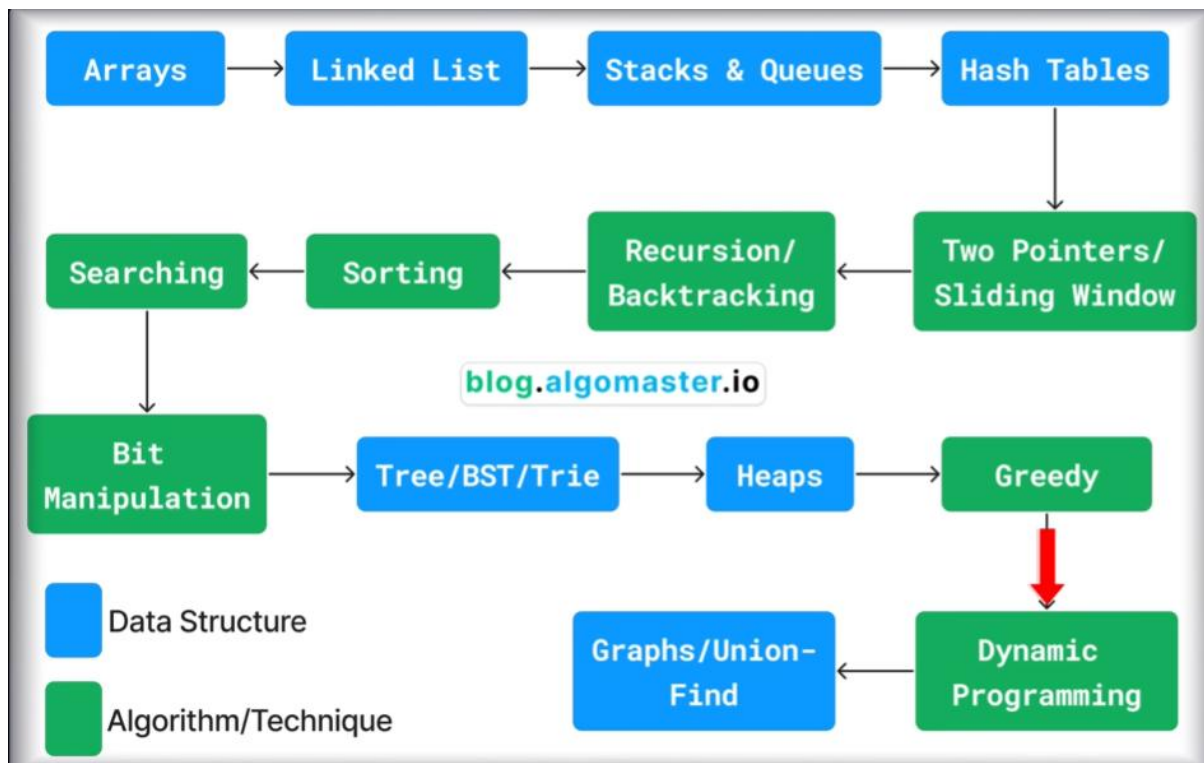




DSA in C++ — Learning Roadmap

Here's a structured roadmap from beginner to interview-ready:

🔧 Phase 1 — C++ Fundamentals (1–2 weeks)

Before DSA, get solid on:

- **STL** — vector, map, set, queue, stack, priority_queue, unordered_map
- **Pointers & References** — essential for linked lists, trees
- **OOP basics** — classes, constructors (useful for custom data structures)
- **Time & Space Complexity** — Big O notation (learn this early, apply everywhere)

📦 Phase 2 — Core Data Structures (4–6 weeks)

Learn each structure: theory → implement from scratch → use STL equivalent

Structure	Key Concepts
Arrays & Strings	Two pointers, sliding window
Linked List	Singly, doubly, fast/slow pointers
Stack & Queue	Monotonic stack, deque

Structure	Key Concepts
Hashing	HashMap, HashSet, collision handling
Trees	BST, traversals (BFS/DFS), height, diameter
Heaps	Min/max heap, <code>priority_queue</code>
Graphs	Adjacency list/matrix, BFS, DFS
Tries	Prefix trees, word search

Phase 3 — Core Algorithms (4–6 weeks)

Algorithm	Topics
Sorting	Merge sort, Quick sort, Counting sort
Binary Search	On arrays, on answer space
Recursion & Backtracking	Subsets, permutations, N-Queens
Dynamic Programming	Memoization → tabulation, classic patterns (knapsack, LCS, LIS)
Greedy	Activity selection, interval scheduling
Graph Algorithms	Dijkstra, Bellman-Ford, Floyd-Warshall, Topological sort, Union-Find

Phase 4 — Advanced Topics (3–4 weeks)

- Segment Trees & BIT (Fenwick Tree)
- Disjoint Set Union (DSU)
- Sliding Window + Two Pointer mastery
- Bit Manipulation
- Advanced DP (bitmask DP, DP on trees)

Practice Strategy

Platforms (in order):

1. **LeetCode** — primary grind (Easy → Medium → Hard)
2. **Codeforces** — competitive programming, sharpens speed
3. **GeeksforGeeks** — great explanations + topic-wise problems

Daily routine:

- 1–2 problems/day minimum
- After solving, read top solutions to learn patterns
- Maintain a **revision sheet** for problems you got wrong

Best Resources

Type	Resource
Video	Striver's A2Z DSA Sheet (YouTube + takeUforward.org) — best for C++
Video	Abdul Bari (algorithms, very clear)
Book	<i>Cracking the Coding Interview</i> (for interviews)
Book	<i>Introduction to Algorithms</i> (CLRS) — for depth
Practice Sheet Striver's SDE Sheet / NeetCode 150	

Timeline Summary

Phase	Duration
C++ + Complexity	1–2 weeks
Data Structures	4–6 weeks
Algorithms	4–6 weeks
Advanced + Mock interviews	3–4 weeks
Total	~4–5 months

Key tip: Don't just read — **code every data structure from scratch at least once** before relying on STL. That's what separates people who understand DSA from those who just memorize it.